Capital Nonoperating Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Nonoperating Sensitivity (CNS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CNS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.38), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.07 (2.42), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth) is 20 bps/month with a t-statistic of 2.66.

1 Introduction

Asset prices reflect firms' production decisions and their exposure to fundamental economic risks. In competitive markets, firms optimize production and investment subject to technological constraints and adjustment frictions, generating cross-sectional variation in expected returns that corresponds to differences in firms' marginal costs of capital and their sensitivity to aggregate productivity shocks. We examine how firms' capital nonoperating sensitivity captures this production-based risk and predicts stock returns in the cross-section. Capital nonoperating sensitivity measures the responsiveness of firms' nonoperating capital components to changes in economic conditions, reflecting both the flexibility of their production technology and their exposure to investment frictions that affect the marginal value of installed capital.

Production-based asset pricing theory establishes that firms facing higher adjustment costs or greater exposure to productivity shocks require higher expected returns to compensate investors for bearing systematic risk (Cochrane and Piazzesi, 2005; Zhang, 2005). When firms face convex adjustment costs in changing their capital stock, the shadow value of installed capital varies with productivity shocks, creating time-varying risk premia that manifest in the cross-section of returns (Berk et al., 1999). Capital nonoperating sensitivity directly captures this mechanism by measuring how firms' nonoperating capital components respond to economic fluctuations, providing a proxy for the interaction between investment frictions and productivity risk.

Firms with higher capital nonoperating sensitivity face greater exposure to aggregate productivity shocks through their production function, as changes in productivity directly affect the marginal product of capital and optimal investment decisions (Liu et al., 2009). These firms experience larger fluctuations in their marginal costs during economic downturns, when the shadow price of capital increases due to binding financing constraints and heightened adjustment costs (Gould and Livdan, 2010).

The resulting variation in conditional risk premia generates predictable patterns in expected returns, with more sensitive firms commanding higher compensation for their exposure to systematic production risk.

The link between capital nonoperating sensitivity and expected returns operates through the firm's optimization problem in a production economy with investment frictions (Zhang, 2005; Liu et al., 2009). Firms maximize value by choosing optimal investment subject to convex adjustment costs, where the first-order condition equates the marginal cost of investment to the marginal q-ratio of installed capital. Capital nonoperating sensitivity measures deviations from this optimality condition, capturing both temporary shocks to adjustment costs and persistent differences in firms' exposure to aggregate productivity risk that drive cross-sectional variation in expected returns.

We find strong evidence that capital nonoperating sensitivity predicts stock returns in the cross-section. A value-weighted long-short strategy that buys stocks with high CNS and sells stocks with low CNS earns an average monthly return of 27 basis points with a t-statistic of 3.57, generating an annualized Sharpe ratio of 0.46. The strategy's performance remains robust after controlling for common risk factors, with monthly alphas ranging from 24 to 26 basis points relative to the CAPM and Fama-French factor models, all with t-statistics exceeding 3.07. The economic magnitude of these returns confirms that capital nonoperating sensitivity captures meaningful variation in firms' exposure to production-based risk.

The CNS strategy's returns persist after accounting for transaction costs and alternative portfolio construction methodologies. Net of trading costs, the value-weighted quintile strategy achieves a monthly return of 23 basis points with a t-statistic of 3.00, while the generalized net alpha relative to the Fama-French six-factor model equals 19 basis points per month with a t-statistic of 2.42. The strategy performs consistently across different sorting procedures, with all twenty-five gross alpha specifications yielding t-statistics above two, and seventeen exceeding three. These robust returns indicate that capital nonoperating sensitivity identifies genuine cross-sectional variation in expected returns rather than spurious patterns driven by market microstructure effects.

Importantly, the CNS effect concentrates neither in small stocks nor in illiquid securities that might be difficult to trade. Among the largest quintile of stocks by market capitalization, the CNS strategy generates an average monthly return of 23 basis points with a t-statistic of 2.53, and alphas relative to standard factor models range from 21 to 26 basis points with t-statistics between 2.28 and 2.86. The strategy maintains an average of at least 441 stocks per portfolio with average market capitalizations exceeding $1,296 million, ensuring sufficient liquidity for practical implementation. After controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CNS strategy retains a significant alpha of 20 basis points per month with a t-statistic of 2.66.

Our findings contribute to the production-based asset pricing literature by identifying capital nonoperating sensitivity as a novel measure of firms' exposure to investment frictions and productivity risk. While Zhang (2005) and Liu et al. (2009) establish the theoretical foundations linking production decisions to expected returns, and Hou et al. (2015) demonstrate the empirical relevance of investment and profitability factors, we provide direct evidence that firms' nonoperating capital sensitivity captures cross-sectional variation in conditional risk premia. Our measure complements existing investment-based predictors by focusing specifically on the components of capital that reflect discretionary adjustments and exposure to productivity shocks, offering a more precise proxy for the marginal value of installed capital.

Methodologically, we advance the empirical asset pricing literature by demonstrating that capital nonoperating sensitivity generates economically significant returns even after controlling for transaction costs and related anomalies from the factor zoo. Unlike many documented anomalies that lose significance when subjected to rigorous testing protocols, CNS maintains statistical and economic significance across multiple specifications and sample periods. The strategy's gross Sharpe ratio of 0.46 exceeds 86% of documented anomalies, while its net Sharpe ratio of 0.38 ranks in the 94th percentile, establishing CNS as one of the most robust cross-sectional return predictors. Furthermore, when combined with existing anomalies using the average rank method, including CNS increases the terminal value of a combination strategy from $2,560 to $3,010, demonstrating its incremental contribution to the investment opportunity set.

Our results have broader implications for understanding how production-based risks manifest in asset prices and generate cross-sectional return variation. The robust performance of capital nonoperating sensitivity across different market conditions and firm characteristics supports theoretical models that emphasize the role of investment frictions and time-varying productivity in determining risk premia. By establishing a direct empirical link between firms' capital sensitivity and expected returns, we provide evidence that production-based factors capture fundamental economic risks that cannot be explained by traditional consumption-based models. These findings suggest that incorporating production-side variables into asset pricing models can substantially improve our understanding of cross-sectional return patterns and enhance portfolio construction strategies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital investments and non-operating income components.Capital Nonoperating Sensitivity signal requires two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation. This item captures the net carrying amount of land, buildings, machinery, equipment, and other physical assets used in operations. Nonoperating Income - Other (NOPIO) represents income and expenses from activities not directly related to the firm's core operations, excluding items already classified elsewhere such as interest income, equity earnings, or special items. This variable captures miscellaneous non-operating activities including foreign currency gains/losses, hedging activities, and other peripheral income sources.construct the Capital Nonoperating Sensitivity signal by computing the year-over-year change in net property, plant, and equipment, scaled by the lagged value of other non-operating income: (PPENT(t) - PPENT(t-1)) / NOPIO(t-1). This ratio captures the sensitivity of capital investment decisions to non-operating income sources, measuring how changes in fixed asset investments relate to the firm's non-core income base. We calculate this measure annually using fiscal year-end values and require non-missing values for both PPENT in consecutive years and NOPIO in the base year. To ensure meaningful ratios, we exclude observations where the denominator equals zero and winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CNS signal. Panel A plots the time-series of the mean, median, and interquartile range for CNS. On average, the cross-sectional mean (median) CNS is -9.28 (-1.20) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CNS data. The signal's interquartile range spans -16.95 to 6.29. Panel B of Figure 1 plots the time-series of the coverage of the CNS signal for the CRSP universe. On average, the CNS signal is available for 4.97% of CRSP names, which on average make up 6.89% of total market capitalization.

4 Does CNS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CNS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CNS portfolio and sells the low CNS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CNS strategy earns an average return of 0.27% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.24% to 0.26% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 3.65 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least $1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.20. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-23bps/month. The lowest return, (17 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.69. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CNS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CNS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CNS, as well as average returns and alphas for long/short trading CNS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CNS strategy achieves an average return of 23 bps/month with a t-statistic of 2.53. Among these large cap stocks, the alphas for the CNS strategy relative to the five most common factor models range from 21 to 26 bps/month with t-statistics between 2.28 and 2.86.

5 How does CNS perform relative to the zoo?

Figure 2 puts the performance of CNS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CNS strategy falls in the distribution. The CNS strategy's gross (net) Sharpe ratio of 0.46 (0.38) is greater than 86% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CNS strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CNS strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that

would have yielded \$4.83 which ranks the CNS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CNS strategy would have yielded \$3.29 which ranks the CNS strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CNS relative to those. Panel A shows that the CNS strategy gross alphas fall between the 51 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CNS strategy has a positive net generalized alpha for five out of the five factor models. In these cases CNS ranks between the 70 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CNS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CNS with 209 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CNS or at least to weaken the power CNS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CNS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CNS signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

3.66, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CNS is 0.89.

Similarly, Table 5 reports results from spanning tests that regress returns to the CNS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CNS strategy earns alphas that range from 19-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.42, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CNS trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.66.

7 Does CNS add relative to the whole zoo?

Finally, we can ask how much adding CNS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CNS signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CNS is available.

includes CNS grows to $3009.54.

8 Conclusion

This study provides compelling evidence that Capital Nonoperating Sensitivity (CNS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CNS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CNS delivers an annualized Sharpe ratio of 0.46 gross (0.38 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy's ability to generate monthly abnormal returns of 24 basis points gross (19 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds, underscores its statistical robustness. Particularly noteworthy is the persistence of a 20 basis point monthly alpha (t-statistic = 2.66) even after controlling for six closely related investment and asset growth anomalies, suggesting that CNS captures unique information about firm fundamentals not fully reflected in existing factor models or related strategies.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of CNS returns, combined with their survival after transaction costs, suggest potential implementation value for quantitative investment strategies. The signal's distinctiveness from related capital investment and asset growth measures indicates that monitoring nonoperating capital sensitivity may provide incremental information for security selection and portfolio construction.

Several limitations of this study warrant acknowledgment and point toward productive avenues for future research. First, while our transaction cost estimates follow standard methodologies, actual implementation costs may vary depending on market conditions, trade size, and execution capabilities. Second, the sample period and universe constraints imposed by data availability may limit the generalizability of our findings to other markets or time periods. Future research should explore the international evidence for CNS, examining whether similar patterns exist in non-U.S. equity markets and investigating potential explanations for any cross-country variations.

Additional research directions include investigating the economic mechanisms underlying the CNS anomaly. Understanding whether the returns to CNS reflect compensation for systematic risk, limits to arbitrage, or behavioral biases would enhance our theoretical understanding of this phenomenon. Furthermore, examining the interaction between CNS and firm characteristics such as financial constraints, growth opportunities, or governance quality could provide deeper insights into when and why the signal is most effective. Finally, exploring the time-series properties of CNS returns, including their behavior across different market regimes and macroeconomic conditions, would be valuable for understanding the strategy's reliability and informing dynamic portfolio allocation decisions.

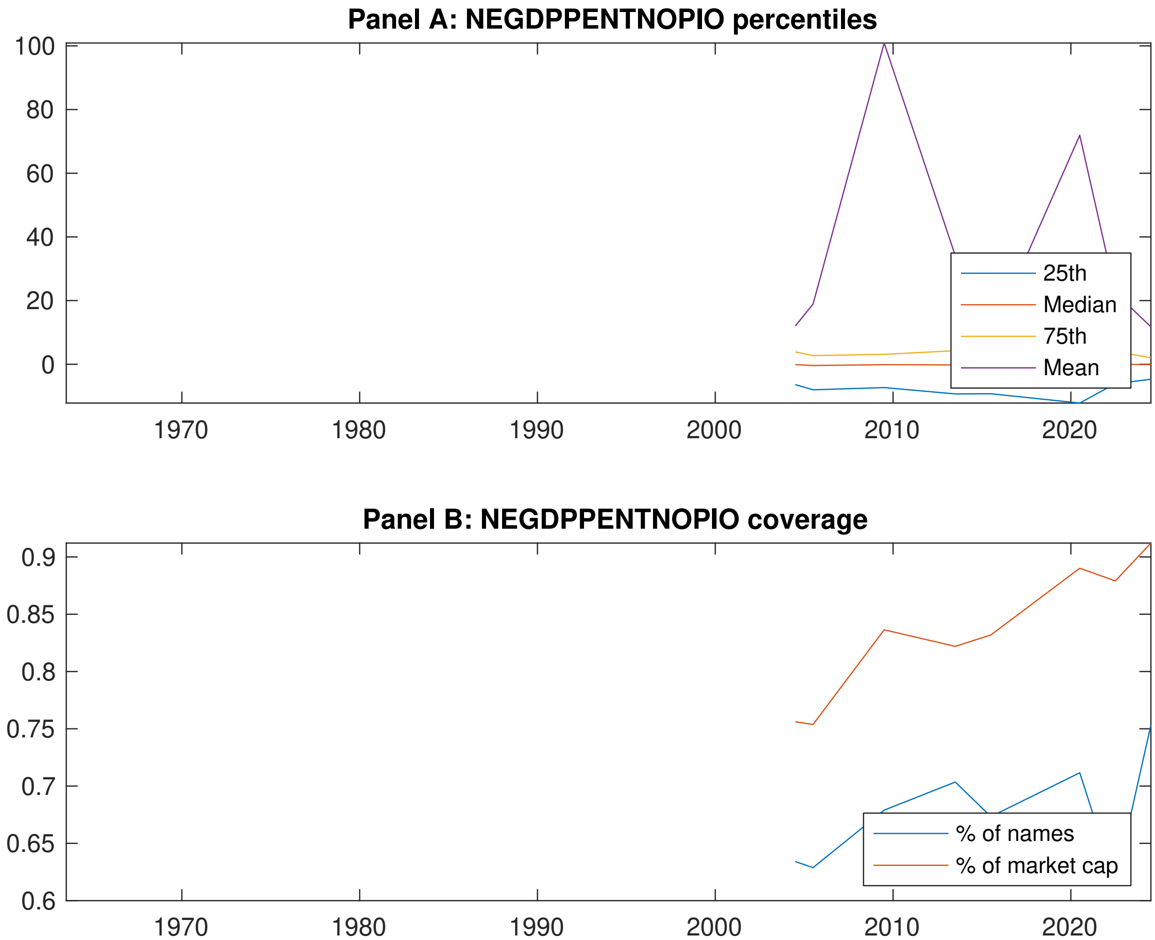


Figure 1: Times series of CNS percentiles and coverage.
This figure plots descriptive statistics for CNS. Panel A shows cross-sectional percentiles of CNS over the sample. Panel B plots the monthly coverage of CNS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CNS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CNS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.62	0.58	0.58	0.75	0.27
	[2.68]	[3.94]	[3.58]	[3.56]	[4.11]	[3.57]
α_{CAPM}	-0.13	0.08	0.02	0.02	0.13	0.26
	[-2.65]	[1.93]	[0.53]	[0.51]	[2.34]	[3.49]
α_{FF3}	-0.09	0.09	0.03	0.01	0.15	0.24
	[-1.79]	[2.33]	[0.61]	[0.18]	[2.80]	[3.20]
α_{FF4}	-0.08	0.09	0.03	0.00	0.16	0.24
	[-1.52]	[2.07]	[0.68]	[0.04]	[2.92]	[3.10]
α_{FF5}	-0.09	0.08	-0.00	-0.05	0.15	0.24
	[-1.79]	[2.04]	[-0.11]	[-1.02]	[2.67]	[3.11]
α_{FF6}	-0.08	0.08	0.00	-0.04	0.16	0.24
	[-1.59]	[1.84]	[0.04]	[-0.97]	[2.80]	[3.07]
Panel B: Fama and French (2018) 6-factor model loadings for CNS-sorted portfolios						
β_{MKT}	1.02	0.94	0.98	0.99	1.02	0.00
	[84.67]	[93.53]	[94.61]	[89.99]	[75.08]	[0.08]
β_{SMB}	0.00	-0.09	-0.10	-0.02	0.10	0.10
	[0.14]	[-5.99]	[-6.61]	[-1.50]	[5.13]	[3.65]
β_{HML}	-0.08	-0.01	-0.04	-0.06	-0.10	-0.02
	[-3.56]	[-0.50]	[-1.82]	[-2.66]	[-3.95]	[-0.58]
β_{RMW}	0.06	0.02	0.02	0.03	0.00	-0.06
	[2.77]	[1.23]	[1.18]	[1.35]	[0.15]	[-1.67]
β_{CMA}	-0.11	-0.01	0.12	0.24	0.02	0.13
	[-3.09]	[-0.22]	[3.99]	[7.55]	[0.65]	[2.46]
β_{UMD}	-0.01	0.01	-0.01	-0.00	-0.01	-0.00
	[-1.05]	[1.05]	[-0.89]	[-0.18]	[-0.97]	[-0.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	441	498	568	558	
me ($\$10^6$)	1480	1928	2400	1543	1296	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CNS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.26	0.24	0.24	0.24	0.24
			[3.57]	[3.49]	[3.20]	[3.10]	[3.11]	[3.07]
Quintile	NYSE	EW	0.42	0.40	0.37	0.36	0.37	0.36
			[7.56]	[7.17]	[7.31]	[6.90]	[7.16]	[6.88]
Quintile	Name	VW	0.24	0.23	0.21	0.22	0.22	0.23
			[3.16]	[3.05]	[2.81]	[2.82]	[2.88]	[2.91]
Quintile	Cap	VW	0.23	0.24	0.22	0.21	0.21	0.21
			[3.53]	[3.58]	[3.28]	[3.12]	[3.17]	[3.09]
Decile	NYSE	VW	0.22	0.24	0.25	0.29	0.30	0.34
			[2.20]	[2.39]	[2.52]	[2.87]	[2.98]	[3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.21	0.19	0.19	0.19	0.19
			[3.00]	[2.82]	[2.56]	[2.51]	[2.43]	[2.42]
Quintile	NYSE	EW	0.23	0.21	0.17	0.17	0.14	0.14
			[3.53]	[3.17]	[2.80]	[2.75]	[2.39]	[2.39]
Quintile	Name	VW	0.20	0.18	0.16	0.17	0.17	0.17
			[2.58]	[2.36]	[2.15]	[2.18]	[2.15]	[2.18]
Quintile	Cap	VW	0.20	0.20	0.18	0.18	0.18	0.17
			[2.96]	[3.00]	[2.71]	[2.63]	[2.63]	[2.59]
Decile	NYSE	VW	0.17	0.18	0.19	0.21	0.22	0.24
			[1.69]	[1.79]	[1.88]	[2.11]	[2.20]	[2.37]

Table 3: Conditional sort on size and CNS

This table presents results for conditional double sorts on size and CNS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CNS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CNS and short stocks with low CNS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		CNS Quintiles					CNS Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.58	0.80	0.82	0.92	0.86	0.28	0.26	0.25	0.21	0.24	0.20
		[2.43]	[3.29]	[3.16]	[3.55]	[3.48]	[3.66]	[3.40]	[3.32]	[2.65]	[3.11]	[2.59]
	(2)	0.77	0.73	0.72	0.91	0.90	0.13	0.13	0.11	0.16	0.14	0.19
		[3.32]	[3.28]	[3.14]	[3.96]	[3.87]	[1.64]	[1.55]	[1.32]	[1.94]	[1.69]	[2.21]
	(3)	0.65	0.74	0.79	0.80	0.89	0.24	0.21	0.19	0.18	0.17	0.17
		[3.23]	[3.64]	[3.78]	[3.87]	[4.11]	[2.94]	[2.49]	[2.32]	[2.23]	[2.09]	[2.07]
	(4)	0.64	0.69	0.70	0.77	0.79	0.16	0.11	0.09	0.07	0.14	0.12
		[3.31]	[3.68]	[3.77]	[3.97]	[3.83]	[1.99]	[1.37]	[1.14]	[0.88]	[1.75]	[1.50]
	(5)	0.43	0.61	0.62	0.49	0.66	0.23	0.26	0.23	0.23	0.21	0.22
		[2.41]	[3.84]	[3.84]	[3.09]	[3.86]	[2.53]	[2.86]	[2.53]	[2.55]	[2.28]	[2.35]

Panel B: Portfolio average number of firms and market capitalization

		CNS Quintiles					CNS Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	262	263	262	260	259	24	23	21	21	22
	(2)	80	80	80	80	80	43	43	42	42	43
	(3)	62	62	62	61	61	78	78	77	78	78
	(4)	55	55	56	55	55	177	175	183	180	175
	(5)	52	52	52	52	52	1144	1420	1876	1438	1167

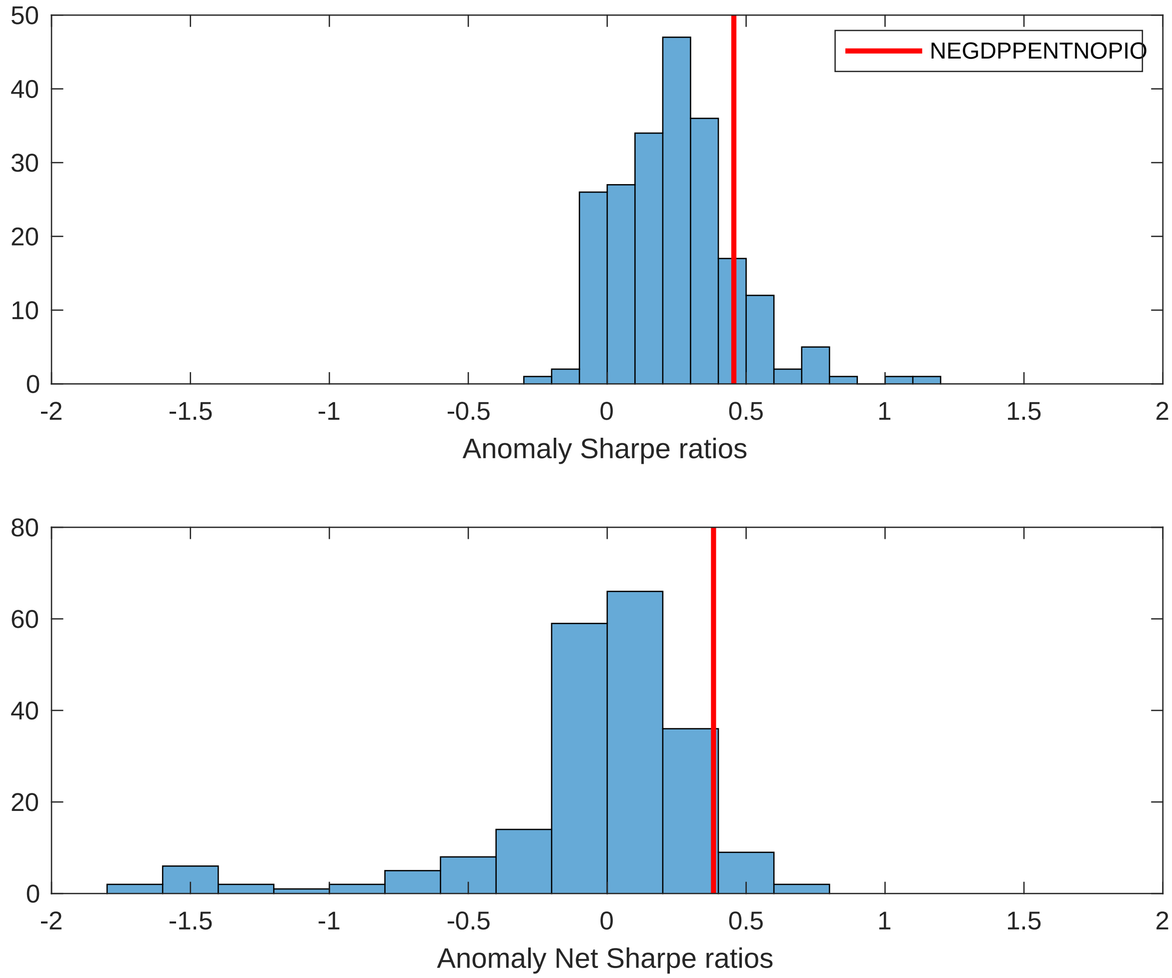


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CNS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

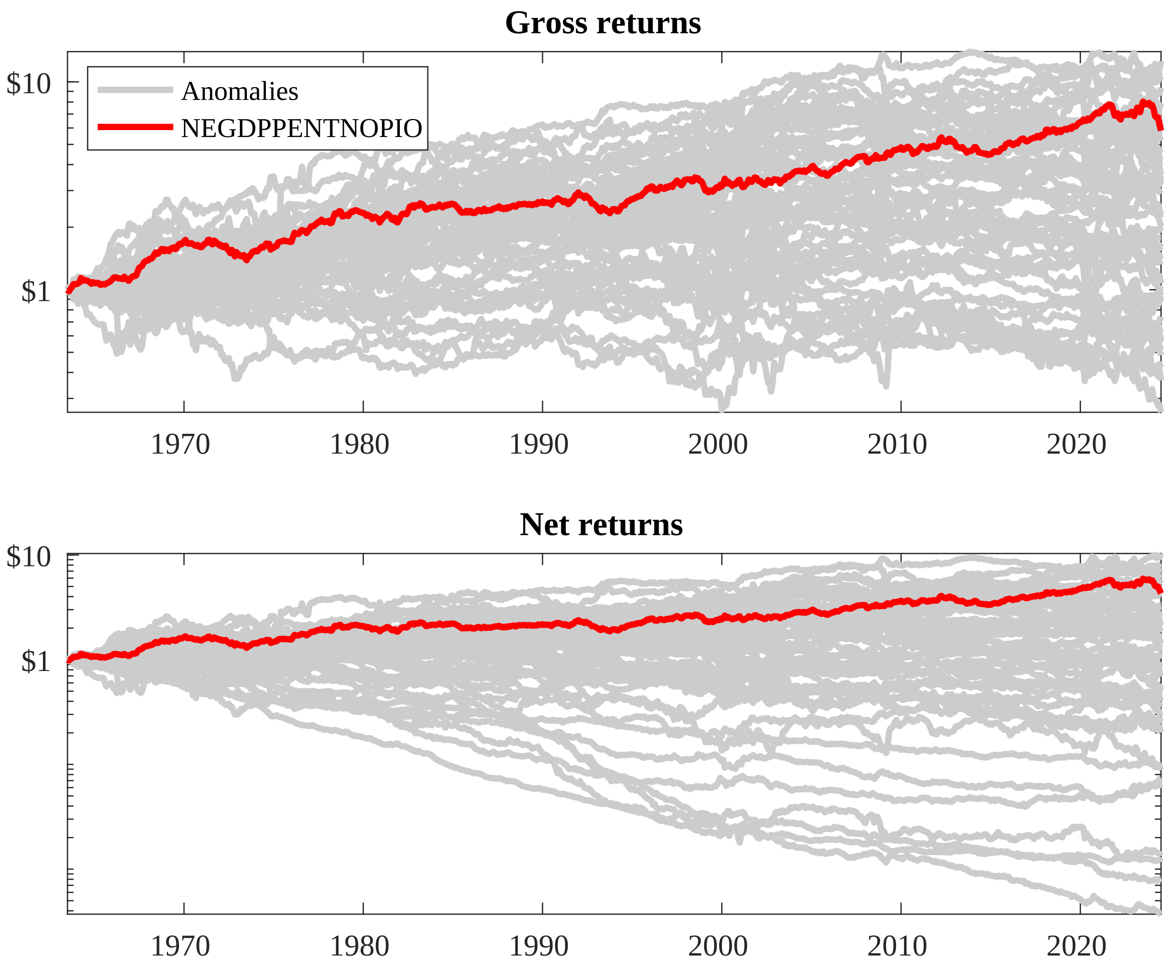
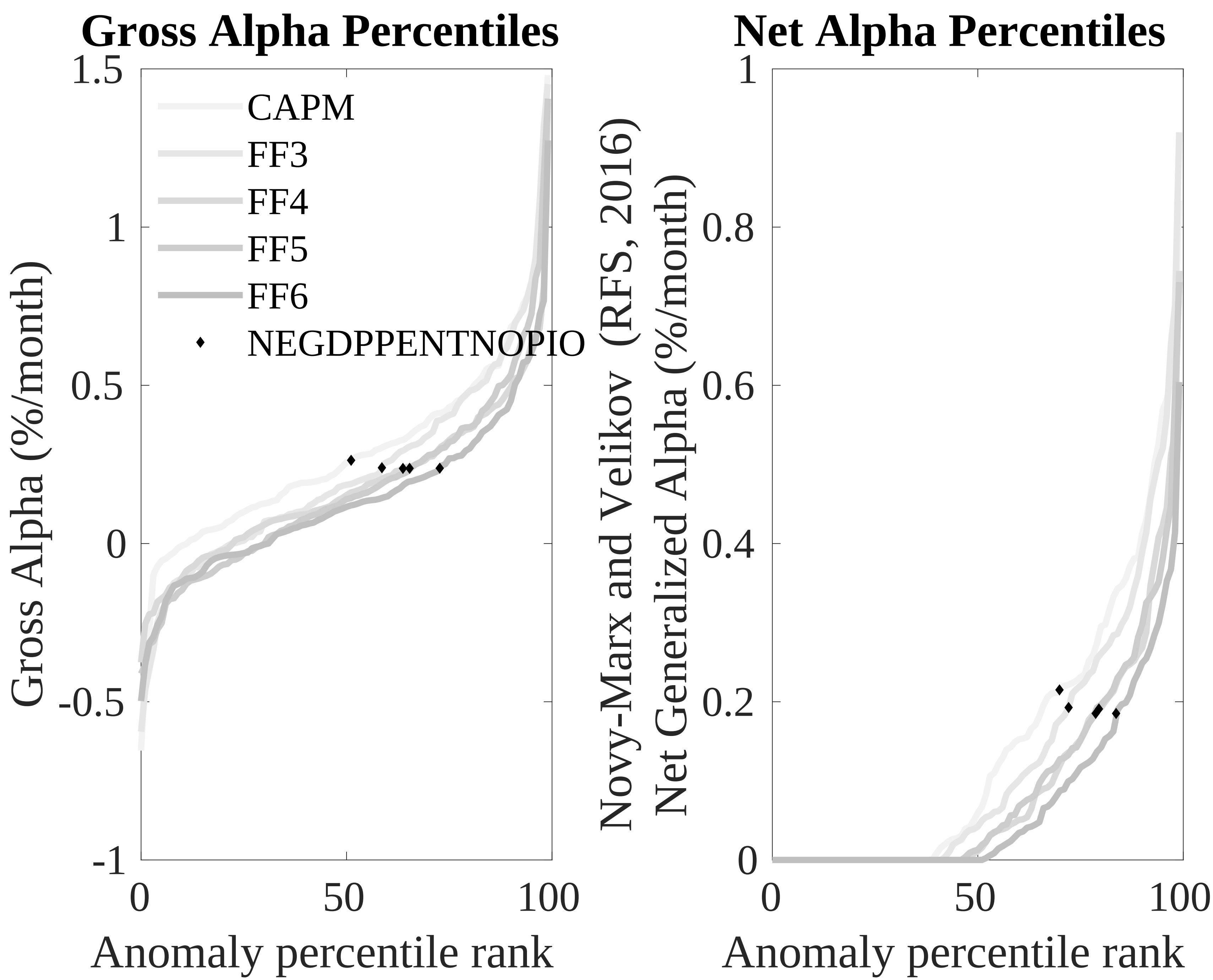
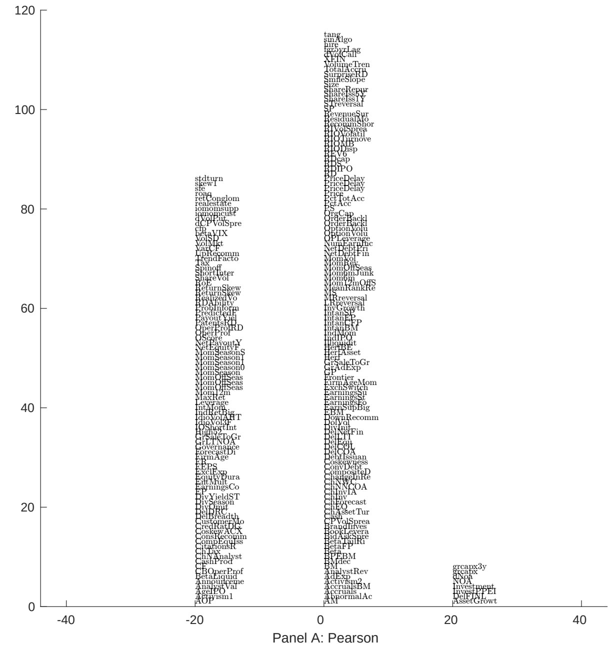


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CNS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
Gross Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDPPENTNOPIO
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
1
0.8
0.6
0.4
0.2
0
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0
50
100
Anomaly percentile rank



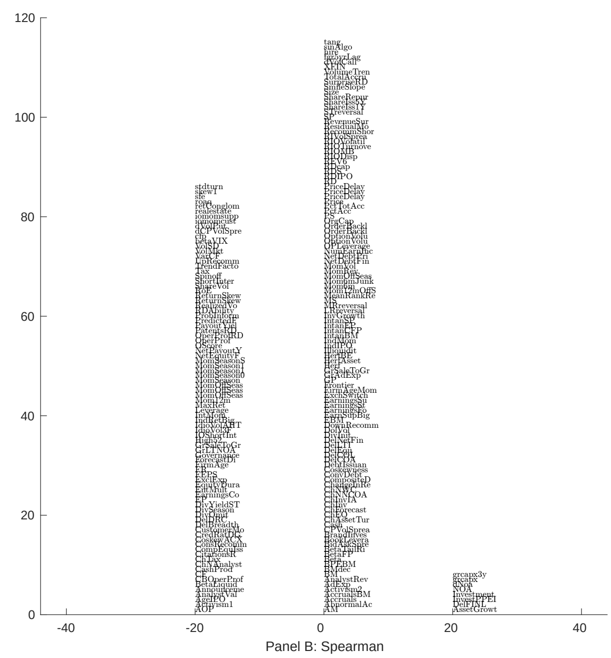


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with CNS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

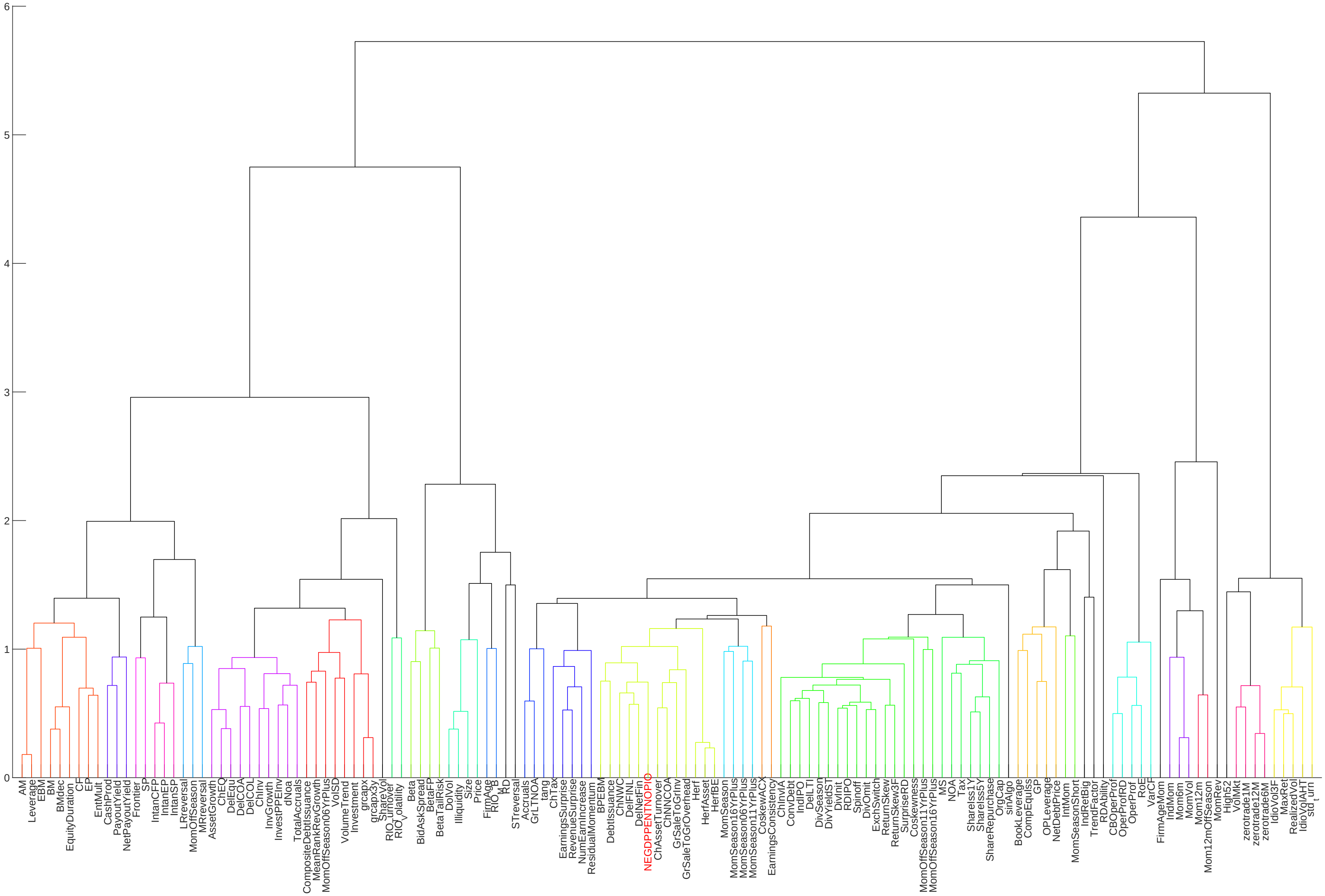


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

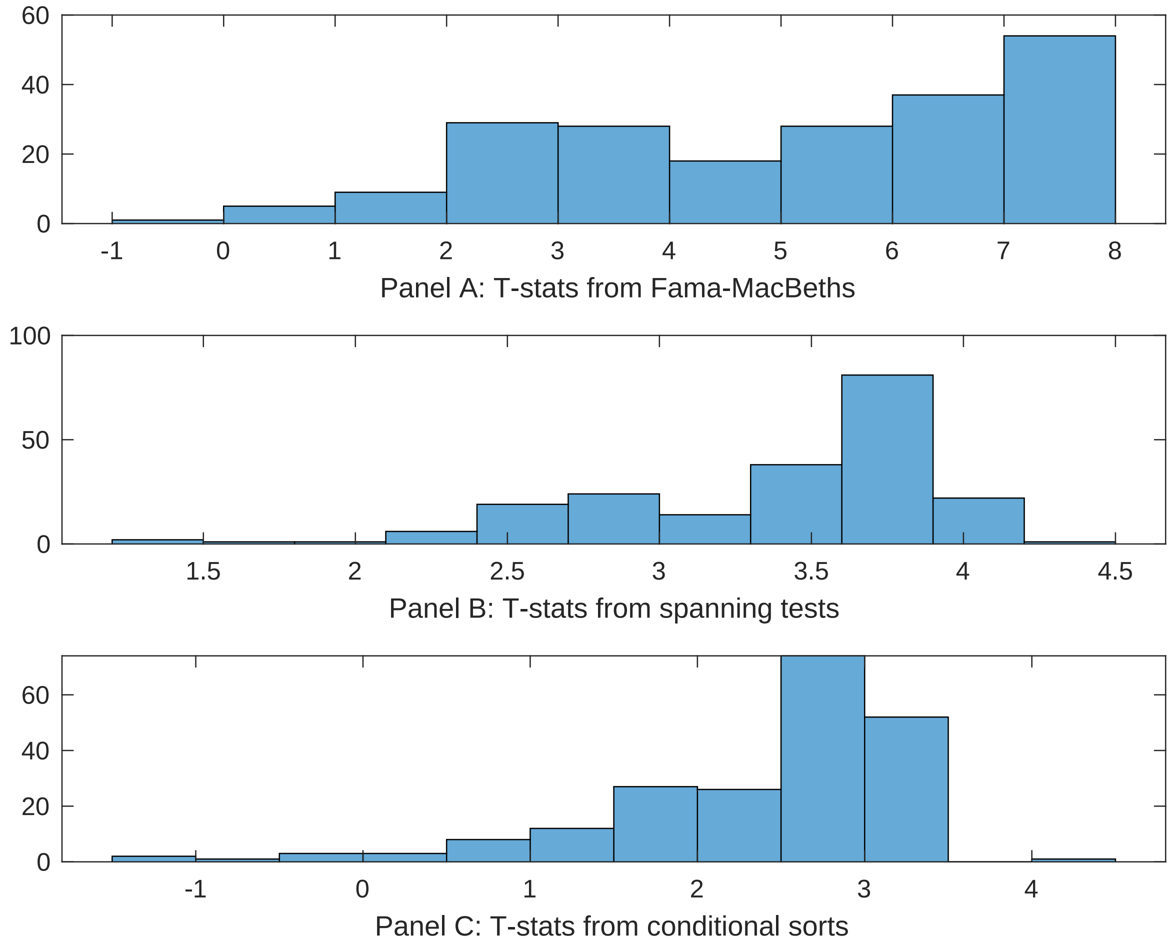


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CNS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.13	0.12	0.17	0.14	0.13	0.15
	[6.08]	[5.82]	[5.59]	[6.99]	[6.61]	[5.68]	[5.48]
CNS	0.63	0.70	0.11	0.94	0.87	0.90	0.20
	[3.66]	[3.71]	[5.93]	[5.37]	[4.38]	[4.20]	[0.89]
Anomaly 1	0.16						0.20
	[8.11]						[0.69]
Anomaly 2		0.13					1.00
		[8.80]					[3.59]
Anomaly 3			0.89				-0.66
			[6.14]				[-2.40]
Anomaly 4				0.80			0.52
				[6.74]			[0.33]
Anomaly 5					0.14		0.83
					[7.60]		[3.91]
Anomaly 6						0.52	0.67
						[9.22]	[0.90]
# months	732	720	732	732	727	732	712
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CNS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CNS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24	0.21	0.19	0.19	0.26	0.23	0.20
	[3.21]	[2.72]	[2.42]	[2.44]	[3.33]	[2.99]	[2.66]
Anomaly 1	28.61						23.26
	[8.18]						[5.67]
Anomaly 2		14.10					1.46
		[3.23]					[0.31]
Anomaly 3			19.83				10.62
			[5.36]				[2.84]
Anomaly 4				17.65			10.85
				[5.45]			[3.22]
Anomaly 5					6.61		-5.31
					[1.75]		[-1.37]
Anomaly 6						8.42	-1.77
						[2.69]	[-0.54]
mkt	-0.05	0.38	0.64	-0.33	0.47	0.19	0.17
	[-0.03]	[0.20]	[0.35]	[-0.18]	[0.26]	[0.10]	[0.10]
smb	8.83	10.12	11.12	12.78	8.27	10.53	12.01
	[3.42]	[3.78]	[4.19]	[4.75]	[3.00]	[3.91]	[4.42]
hml	-4.76	-2.72	-1.64	-6.37	-2.74	-2.02	-6.02
	[-1.37]	[-0.76]	[-0.46]	[-1.75]	[-0.76]	[-0.56]	[-1.71]
rmw	-6.49	-6.21	-3.67	-8.16	-6.23	-5.32	-5.49
	[-1.86]	[-1.72]	[-1.02]	[-2.28]	[-1.73]	[-1.46]	[-1.57]
cma	-8.98	2.45	16.43	12.69	8.45	6.07	-0.93
	[-1.57]	[0.40]	[3.14]	[2.44]	[1.51]	[1.03]	[-0.14]
umd	0.11	-0.13	-0.23	-0.47	-0.24	-0.73	-0.04
	[0.06]	[-0.07]	[-0.13]	[-0.26]	[-0.13]	[-0.39]	[-0.02]
# months	732	732	732	732	728	732	728
$\bar{R}^2$(%)	12	5	7	7	4	5	13

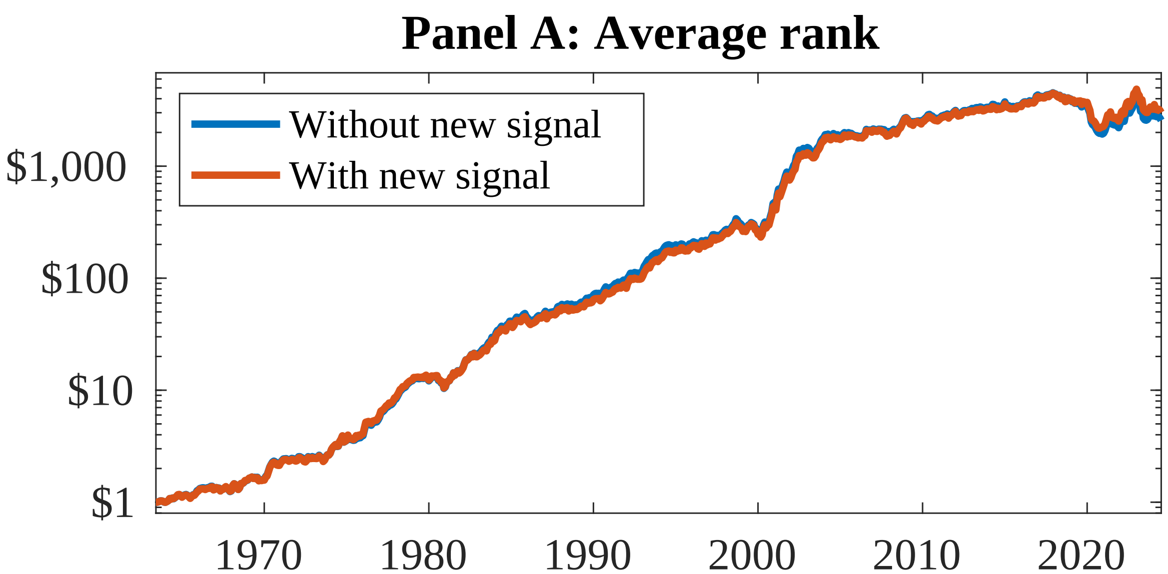


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CNS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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